

Project Design Phase-II

Technology Stack (Architecture & Stack)

Attribute	Details
Date	08 March 2026
Team ID	
Project Name	VoltEdge: EV Market Analytics Portal
Maximum Marks	4 Marks

Technical Architecture

The VoltEdge architecture follows a **3-Tier Client-Server model**. Data flows from a relational database, is processed through a visualization engine, and is served via a lightweight web framework.

Table-1: Components & Technologies

S.No	Component	Description	Technology
1.	User Interface	Web-based interactive dashboard and information portal.	HTML5, CSS3 (VoltEdge Dark Theme), JavaScript
2.	Application Logic-1	Backend server handling routing, request management, and data rendering.	Python (Flask Framework)
3.	Application Logic-2	Business intelligence engine for data processing and aggregation.	Tableau Engine / VizQL
4.	Application Logic-3	Data connectivity and query execution.	SQL / Psycopg2 (PostgreSQL adapter)
5.	Database	Relational data storage for EV specifications and regional stats.	PostgreSQL (pgAdmin 4)
6.	Cloud Database	Cloud storage for hosted analytics (Tableau Public).	Tableau Hyper Extract
7.	File Storage	Storage for local CSV datasets and project assets.	Local Filesystem / GitHub
8.	External API-1	Embedding and interaction with live visualization assets.	Tableau JavaScript API

S.No	Component	Description	Technology
9.	External API-2	Mapping and geospatial data rendering.	OpenStreetMap (via Tableau)
10.	ML Model	Predictive logic for Efficiency vs Price (Planned).	Python Scikit-learn (Linear Regression)
11.	Infrastructure	Application development on local environment and cloud hosting.	Local Server (Development), Tableau Public (Deployment)

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Utilizes community-driven frameworks for server and database management.	Flask (BSD License), PostgreSQL (PostgreSQL License)
2.	Security Implementations	Access control for database management and secure embedding.	SQL Role-Based Access Control (RBAC), HTTPS/SSL for hosted Viz
3.	Scalable Architecture	3-Tier architecture allows independent scaling of the DB and UI layers.	Modular Python/Flask Architecture
4.	Availability	High availability through cloud-hosted visualization servers.	Tableau Public Cloud (High Uptime)
5.	Performance	Optimized query response using data snapshots to reduce latency.	Tableau Data Extracts (.hyper) for < 2s load times

References

- **Architecture:** [IBM Cloud Architecture Center](#)
- **Diagramming:** [C4 Model for Visualising Software Architecture](#)
- **Platform:** [Tableau Public Governance](#)